CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 3: Compound Interest (Using Formula)

Topic-wise short notes with solved examples

The compound interest formula and its variations: different rates, half-yearly compounding, fractional periods, finding the principal, rate or time, and applications to growth and depreciation.

1 The formula

- If a principal P is invested at $r\%$ per annum compound interest for n years, compounded yearly, the amount A is

$$A = P\left(1 + \frac{r}{100}\right)^n$$

- Compound interest $= A - P = P\left[\left(1 + \frac{r}{100}\right)^n - 1\right]$.
- Each year the amount is multiplied by the same factor $\left(1 + \frac{r}{100}\right)$.

Example 1.1 Find the amount and the C.I. on ₹8000 for 3 years at 5% per annum.

SOLUTION

$$A = 8000\left(\frac{21}{20}\right)^3 = 8000 \times \frac{9261}{8000}$$

Amount ₹9261; C.I. ₹1261

Example 1.2 Find the C.I. on ₹25,000 for 2 years at 12% per annum.

SOLUTION

$$A = 25000 \times (1.12)^2 = 25000 \times 1.2544 = 31360$$

₹6360

2 Different rates in different years

- If the rates for successive years are $r_1\%$, $r_2\%$, $r_3\%$:

$$A = P\left(1 + \frac{r_1}{100}\right)\left(1 + \frac{r_2}{100}\right)\left(1 + \frac{r_3}{100}\right)$$

Example 2.1 Find the amount on ₹10,000 for 3 years if the rates are 5%, 8% and 10%.

SOLUTION

$10000 \times 1.05 \times 1.08 \times 1.10$

₹12,474

3 Other compounding periods

- **Half-yearly**: rate $\frac{r}{2}\%$ per period and $2n$ periods: $A = P\left(1 + \frac{r}{200}\right)^{2n}$.
- **Quarterly**: $A = P\left(1 + \frac{r}{400}\right)^{4n}$.
- **Time not a whole number of years** (compounded yearly), such as $2\frac{1}{2}$ years: use the formula for the whole years and simple interest for the remaining fraction.

$$A = P\left(1 + \frac{r}{100}\right)^{2}\left(1 + \frac{\frac{1}{2} \times r}{100}\right)$$

Example 3.1 Find the amount on ₹16,000 for $1\frac{1}{2}$ years at 10% per annum, compounded half-yearly.

SOLUTION

$$A = 16000\left(1 + \frac{5}{100}\right)^{3} = 16000 \times \frac{9261}{8000}$$

₹18,522

Example 3.2 Find the amount on ₹20,000 for $2\frac{1}{2}$ years at 10% per annum, compounded yearly.

SOLUTION

$$20000 \times (1.1)^2 \times \left(1 + \frac{5}{100}\right) = 24200 \times 1.05$$

₹25,410

4 Finding P, r or n

- **Principal**: $P = A \div \left(1 + \frac{r}{100}\right)^{n}$.
- **Rate**: find $\left(1 + \frac{r}{100}\right)^{n} = \frac{A}{P}$ and take the nth root.
- **Time**: write $\frac{A}{P}$ as a power of $\left(1 + \frac{r}{100}\right)$ and compare the exponents.

Example 4.1 What sum amounts to ₹6655 in 3 years at 10% per annum compound interest?

SOLUTION

$P = 6655 \div (1.1)^3 = 6655 \div 1.331$

₹5000

Example 4.2 At what rate will ₹4000 amount to ₹4840 in 2 years, compounded yearly?

SOLUTION

$$\left(1+\frac{r}{100}\right)^2=\frac{4840}{4000}=\frac{121}{100}=\left(\frac{11}{10}\right)^2$$

10% per annum

Example 4.3 In how many years will ₹8000 amount to ₹9261 at 5% per annum?

SOLUTION

$$\left(\frac{21}{20}\right)^n=\frac{9261}{8000}=\left(\frac{21}{20}\right)^3$$

3 years

Example 4.4 A sum becomes $\frac{27}{8}$ times itself in 3 years at compound interest. Find the rate.

SOLUTION

$\left(1+\frac{r}{100}\right)^3=\left(\frac{3}{2}\right)^3$, so $\frac{r}{100}=\frac{1}{2}$

50% per annum

5 C.I. and S.I. compared

- For **2 years**: C.I. $-$ S.I. $=P\left(\frac{r}{100}\right)^2$.
- For **3 years**: C.I. $-$ S.I. $=P\left(\frac{r}{100}\right)^2\left(3+\frac{r}{100}\right)$.

Example 5.1 The difference between the C.I. and the S.I. on a sum for 2 years at 5% is ₹20. Find the sum.

SOLUTION

$P\times\left(\frac{5}{100}\right)^2=20$, so $P=20\times400$

₹8000

Example 5.2 Find C.I. $-$ S.I. on ₹8000 for 3 years at 5%.

SOLUTION

$$8000\times\frac{1}{400}\times3.05$$

₹61

6 Growth and depreciation

- **Growth** at $r\%$ per year: value after n years $= V_0\left(1+\frac{r}{100}\right)^n$.
- **Depreciation** at $r\%$ per year: value after n years $= V_0\left(1-\frac{r}{100}\right)^n$.
- **Value n years ago** (growth): $V_0 \div \left(1+\frac{r}{100}\right)^n$.
- Growth one year and decline the next: multiply the two factors.

Example 6.1 The population of a town is 64,000 and grows at 5% per year. Find the population after 3 years.

SOLUTION

$64000 \times \left(\frac{21}{20}\right)^3$

74,088

Example 6.2 A machine depreciates at 10% per year and is now worth ₹72,900. What was it worth 3 years ago?

SOLUTION

$V \times (0.9)^3 = 72900$, so $V = 72900 \div 0.729$

₹1,00,000

Example 6.3 The population of a village was 20,000. It rose by 10% in the first year and fell by 10% in the second. Find the population after 2 years.

SOLUTION

$20000 \times 1.1 \times 0.9$

19,800

7 Common mistakes

- Using n instead of $2n$, or r instead of $\frac{r}{2}$, for half-yearly compounding.
- Writing the factor as $(1+r)$ instead of $\left(1+\frac{r}{100}\right)$.
- Using a plus sign for depreciation.
- Thinking that a rise of 10% followed by a fall of 10% gives the original value. It gives 99% of it.
- Reporting A when C.I. is asked.
- For $2\frac{1}{2}$ years, using the power 2.5 instead of two full years and simple interest for the half year.